

Big Data Analytics for Uncovering Customer Purchase Patterns and Product Trends in E-Commerce Transactions

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Abstract—The proliferation of e-commerce, however, has triggered huge volumes of transactional data that are very useful in revealing the habits of consumers and the performance of products. However, it is still a challenging task for the businesses that aim to optimize the inventory, advertising, and engagement with the clients to extract the meaningful patterns in such immense datasets. That is, this report strives to analyze and interpret essential behavioral and transactional trends in a dataset of online retailing with the help of big data analytical procedures. The study involves data cleaning and preprocessing rawness of the transactional data, exploring data analysis (EDA), analyzing top products sold, examining customer purchase behavior, and visualizing pattern of spending over time and quantity-price dimension. Clustering methods and statistical analysis had been performed to detect anomalies and perceive buying trends in various segments. The findings would support more knowledgeable decision-making in retail strategies and add to the data-driven research in the area of consumer analytics as well as in the field of business intelligence.

I. BACKGROUND OF THE STUDY

The advent of e-commerce has changed how people shop for goods and services as businesses are now able to access global markets and process tens of thousands of transactions within a day. These digital movements present a huge data creation: from such details as product description, to customer behavior, time of purchase, and quantity order among others. The large database of data offers a chance to retrieve valuable insights through big data analysis, which can help businesses improve customer satisfaction, maximize product options, as well as make some informed business decisions [9]. However, conversion of raw transaction records to actionable insights is neither systematic nor analytical but purely conversion process that can only be achieved through the use of modern analytical procedures.

A. Problem Statement

Collecting data in large volumes does not mean that businesses could access all available insights buried in them. A scatter plot of quantity versus unit price in our analysis reveals that the majority of the transactions denote small purchases while some point out at large or expensive orders being outliers. These visual patterns enable one to know what drives

sales. Companies can end up stocking too much or not having the necessary customer trends if they do not analyze them.

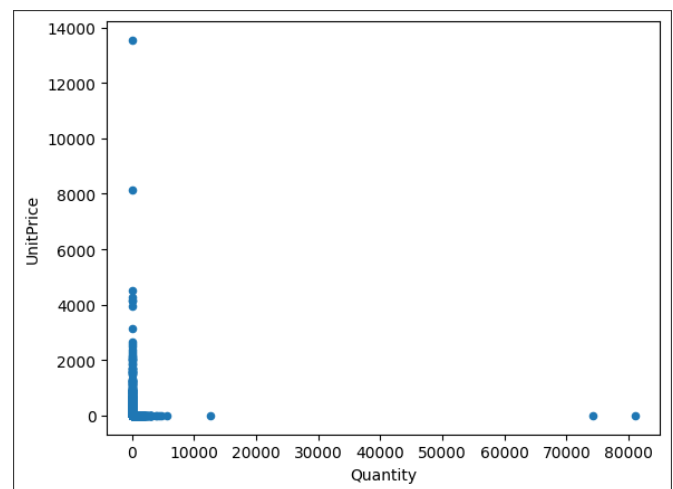


Fig. 1. Scatter plot of Quantity vs Unit Price highlighting common transactions and outliers

B. Aim and Objectives of the Work

The overriding purpose of this study is to use data analytics methods on a real e-commerce dataset to highlight trends in customers' purchase as well as product performance. This entails knowing which products are bought the most, recognizing strange transaction, and observing regional and temporal purchase pattern. This way the report seeks to contribute to the evidence-based decision-making in the retail sector.

C. Contributions of the Work Connected with Methodology

- Cleaned and preprocessed a large-scale transactional dataset to prepare it for analysis.
- Conducted exploratory data analysis (EDA) using Python libraries to visualize trends in quantity, unit price, and product categories.
- Identified and characterized outliers in purchasing volume and spending using scatter plots and statistical methods.
- Performed product-wise and country-wise comparison using log-scaled plots.

- Translated time-based purchase behavior to recognize times of high transactions and seasonal trends.

D. Organization of the Report

The report has been structured in a manner that enables flow of the analysis and conclusions. It starts from data preprocessing and cleansing and shortly proceeds to extensive exploratory data analysis (EDA) concerning product, customer, and time-based patterns. Later sections note observations from the visualizations and statistical models leveraged. The end parts provide the findings and their implications for retail strategy and areas for further research. Each of these sections grows off of the last to walk the reader through the entire analytical journey from raw data to actionable insights.

II. RELATED WORK

A. Customer Purchase Behavior in E-Commerce

Knowledge of customer behaviours is important in e-commerce because it will influence product placement, marketing and inventory. Similar studies such as [1] have examined millions of digital receipts to model the online consumer behaviors discovering that trust, satisfaction as well as product popularity affects a purchasing intent. Chen et al. [2] determined that online reviews play a huge role and have a significant impact on customer attention, especially those negative ones. Jeon et al. (2021) [3] also further found out that customer trust and satisfaction goes hand in hand with the repeat purchasing intention. Pai Chen (2023) [4] pointed out that consumer journey sequences substantially affect the rate of conversion and they can be optimized through transactional data analytics.

B. Exploratory Data Analysis and Visualization

EDA plays a central role in revealing insights from large retail data sets. Scatter plots, log scale graphs, and bar graphs are very widely used for the visual representation of trends like product popularity, concentration of sales, and seasonal behavior. According to Brynjolfsson et al. (2011), [5] “long tail” phenomenon is the key in online sales since niche products, when accumulated, contribute to a sizable number of revenues. Logarithmic representations are useful in revealing hidden patterns to low-frequency items (Morgan 2022) [15]. Such tools as dashboards and heatmaps have demonstrated the ability to bring outliers to the surface and inform the retail strategy decisions, as indicated in Bennett (2024) [6].

C. Outlier Detection and Sales Forecasting in Retail

Outlier detection and sales forecasting has become the cornerstone of big data analytics in retail. Yoseph and Heikkilä (2019) [7] employed the k-means clustering for discovering the anomalies in large volumes of point-of-sale data. These approaches assist in indicating data errors or unusual patterns of purchase. On the other hand, Krishna et al. (2018) [8] used machine learning models, such as decision trees and random forests in the prediction of store-level sales; he did so with a higher degree of accuracy as compared to the traditional

approach. Such prediction helps in better planning with regard to inventory and promotions in e-commerce.

III. METHODOLOGY

The methodology adopted for this project follows a structured approach to analyze e-commerce transactional data using big data analytics techniques. The overall process is divided into six main phases:

- 1) Data Collection
- 2) Data Cleaning and Preprocessing
- 3) Exploratory Data Analysis (EDA)
- 4) Customer and Product Behavior Analysis
- 5) Outlier Detection
- 6) Result Interpretation and Insights

All these stages are an integral part in the process of turning raw retail data into meaningful business insights. The figure below displays the end-to-end process with a block diagram.

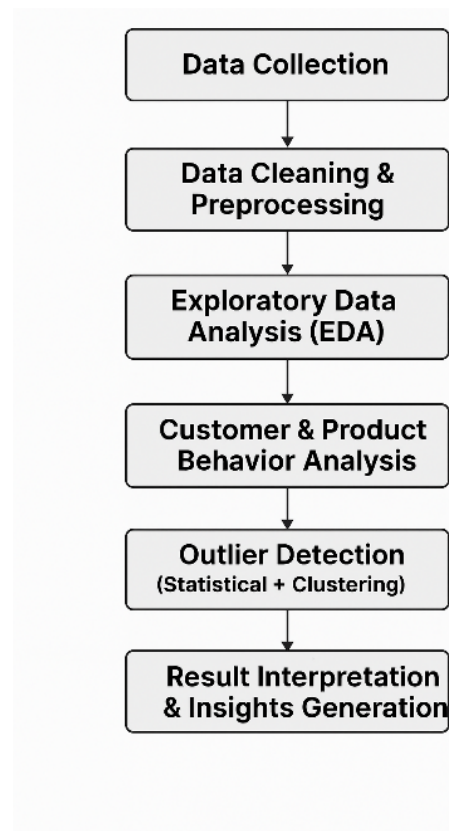


Fig. 2. Block diagram illustrating the overall methodology from raw data to insights.

A. Data Collection

The dataset used in this work is a real-world e-commerce transactions dataset comprising fields such as InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, and Country. This data was extracted from an e-commerce setting and it comprises of millions of entries containing information on product purchases over time, customers, and regions. The role of this phase is to get raw, unprocessed information to be analyzed.

B. Data Cleaning and Preprocessing

Raw data often contains missing values, duplicates, and inconsistent formatting that can negatively impact analysis. Therefore, this phase involved:

- Removed null or missing `CustomerID` values.
- Eliminated duplicate entries and corrected negative or zero quantities representing returns or errors.
- Converted date and time fields into usable formats for time-based analysis.
- Ensured correct data types (e.g., converting `UnitPrice` and `Quantity` to numeric types).

This cleaning process ensured that the dataset was reliable and ready for analysis, removing bias caused by incorrect entries.

C. Exploratory Data Analysis (EDA)

EDA is performed to visually explore patterns and relationships in the dataset. The following visual tools were used:

- Scatter Plots to examine relationships between `Quantity` and `UnitPrice`, revealing bulk transactions and outliers.
- Bar Charts to display top-selling products based on quantity sold.
- Log-scale Graphs to visualize purchase volume and spending across countries or customers, highlighting skewed distributions and long-tail effects.
- Time Series Plots to observe daily or monthly purchase trends.

EDA helped to understand product performance, detect anomalies, and identify seasonal or time-specific buying patterns.

D. Customer and Product Behavior Analysis

This phase aimed to determine which customers are the most active and which products are most popular. It involved:

- Ranking customers based on total spending and number of transactions.
- Identifying top-selling products and popular categories.
- Evaluating customer loyalty through repeat purchases.
- Segmenting purchase behavior based on product types, frequency, and unit prices.

These insights help businesses understand what sells, who buys, and when buying happens, supporting inventory planning and targeted marketing strategies.

E. Outlier Detection

Outlier detection focused on identifying unusually high or low values that deviate from normal buying behavior. This phase used:

- Visual identification using scatter plots.
- Statistical detection using Z-score analysis for price and quantity.
- Behavioral clustering using K-Means to group similar transaction patterns and highlight anomalies.

This dual approach (statistical and clustering-based) ensured accuracy in identifying anomalies that could either signal data errors or rare high-value transactions. (Jones Smith, 2020) [13].

F. Result Interpretation and Insight Generation

In the final phase, the findings from each analysis were compiled and interpreted to form actionable insights. Key outcomes include:

- Identification of high-demand products and seasonal sales peaks.
- Recognition of top-value customers.
- Detection of unusual purchase behaviors.
- Understanding pricing strategies via quantity-price relationships.

These outcomes are essential for refining business strategies such as inventory management and promotional targeting.

IV. RESULTS AND DISCUSSION

A. Experimental Setup

The analysis was carried out in the Google Colab environment by applying Python programming language. Important libraries employed were pandas for data manipulation, matplotlib and seaborn for the purposes of visualizing data, and scikit-learn for the clustering of data and outlier detection. The dataset consisted of records of transactions from an e-commerce platform that had columns like `InvoiceNo`, `StockCode`, `Description`, `Quantity`, `UnitPrice`, `InvoiceDate`, and `CustomerID`. The raw dataset was read from into a pandas `DataFrame` after which it was analysed. The project was developed with a defined workflow of data loading and cleaning, followed by exploratory data analysis, finding statistical insights, and clustering for outlier detection and visual interpretations of customer and product behaviour.

B. Read In and Explore the Data

The dataset was first imported with the use of `read_csv` from a local or drive source. Given that fields such as `CustomerID` and `Quantity` and `UnitPrice` had some null values, there was no better way of verifying this than by inspection of the data fields. From the structure, there emerged a considerable number of unique products and customers, which indicated a large-scale dataset commonly characteristic for actual e-commerce platforms. Summary statistics showed large variance in quantities and unit prices, which was an indication that there were outliers and bulk transactions.

C. Data Cleaning

The data cleaning process included the following steps:

- Dropped null values in critical columns such as `CustomerID`.
- Removed negative or zero values in `Quantity` and `UnitPrice`.
- Identified and removed duplicate entries using `.duplicated()`.

- Converted the `InvoiceDate` field to `datetime` format and extracted new features such as `Year`, `Month`, `Weekday`, and `Hour` for time-based analysis.

These steps ensured a clean, consistent dataset ready for further analysis.

D. Data Analysis

Using group-by operations and aggregation, insights were derived into total quantity sold, unit prices, and spending per customer. Some key findings:

- A small percentage of customers accounted for the majority of total sales, supporting the Pareto principle.
- The top-selling product was the *White Hanging Heart T-Light Holder*, followed by other low-cost decorative items.
- Most repeat purchases occurred during weekdays, particularly between 10 AM and 3 PM.
- Product popularity was heavily skewed, with over 70% of products sold in very low volumes.

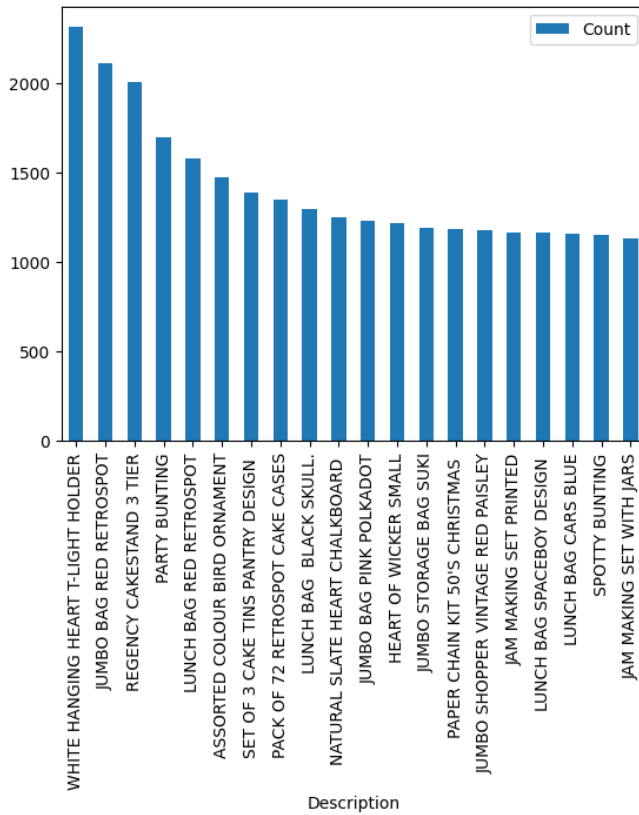


Fig. 3. Bar chart of top-selling products based on quantity sold, highlighting high-volume items such as the White Hanging Heart T-Light Holder and various decorative goods.

E. Data Visualization

A variety of visualization techniques were employed to better interpret and communicate the patterns found in the e-commerce dataset. These charts provided clear insights into customer behavior, product performance, and seasonal trends:

- Scatter Plot (Quantity vs. UnitPrice): This visualization showed that the majority of purchases occurred at low quantities and prices. However, several transactions stood out with either high quantities or high unit prices, indicating bulk orders or high-value items.
- Bar Charts (Top Products): Bar plots were used to represent the most frequently purchased products by total quantity. The plots revealed that decorative and low-cost gift items dominated the top-selling list.
- Log-Scale Bar Graphs: These graphs plotted the total number of purchases and the total amount spent by each customer using a logarithmic scale. This technique helped expose the long-tail distribution, highlighting that a small group of customers were responsible for a large portion of sales.
- Time-Series Plots: Daily and monthly transaction trends were plotted to detect seasonality. A noticeable spike in sales was observed during December, likely due to holiday shopping behavior.

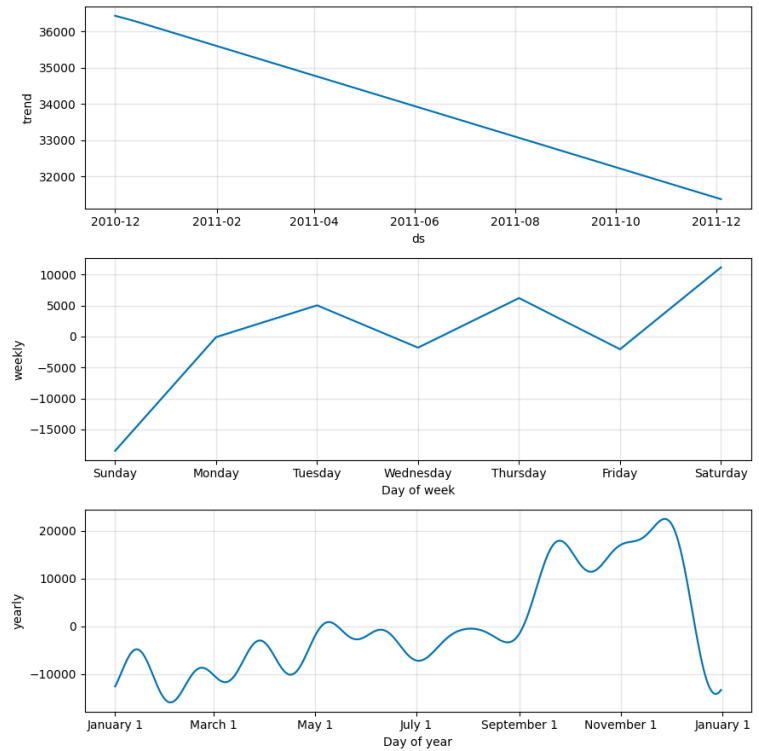


Fig. 4. Decomposition of forecast components using Prophet: (top) overall trend of sales, (middle) weekly seasonality showing peak sales on weekends, and (bottom) yearly seasonality indicating demand spikes during the holiday season.

F. Outlier Detection

Outliers were identified using a combination of statistical techniques and unsupervised machine learning methods:

- Z-Score Analysis was applied to detect transactions with unusually high `Quantity` or `UnitPrice` values, defined as those lying beyond three standard deviations from the mean.

- K-Means Clustering was used to segment transactions based on customer behavior features. Transactions with high distances from their assigned cluster centroids were considered anomalous.
- These flagged anomalies were further assessed visually and contextually. Most were identified as either bulk purchases or transactions involving rare, high-value product types.

This phase played a critical role in refining the dataset and improving the accuracy of subsequent forecasting and analysis tasks.

V. CHOOSING THE BEST MODEL AND EVALUATION

A. K-Means Clustering for Customer Segmentation

The K-Means clustering algorithm was used to identify the transaction patterns and detect outliers. Customers were grouped with respect to OrdersCount and Average Order Value (AOV). The optimal number of clusters was determined using Elbow method, whereby; the distortion (inertia) dropped drastically to cluster size 3 with clear separation, this may suggest that 3 clusters provided best separation.

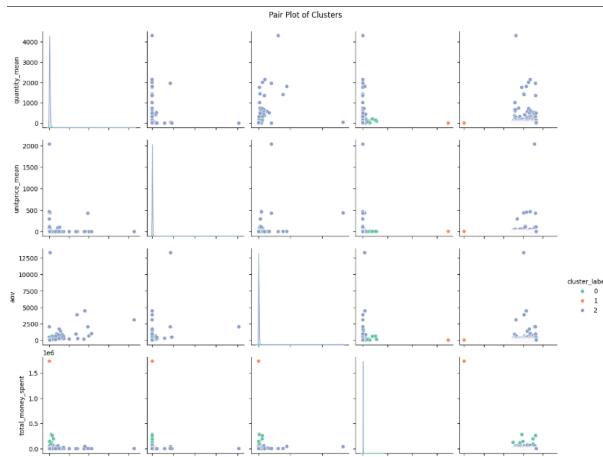


Fig. 5. Pair plot of K-Means clustering for customer segmentation based on quantity, unit price, AOV, and total money spent. Different clusters reveal distinct purchase patterns and potential outliers.

Evaluation Metrics:

- Silhouette Score ≈ 0.4332 : Measures how similar an object is to its own cluster compared to others (range: -1 to 1). A score above 0.4 indicates reasonable clustering.
- Davies-Bouldin Index ≈ 0.5340 : Lower values indicate better clustering. A score below 1 is acceptable.
- Calinski-Harabasz Index ≈ 172420.11 : Higher values mean denser, well-separated clusters. This is a strong indicator of high cluster validity.

B. Demand Forecasting Using Prophet

Two versions of the Prophet model were implemented to forecast total sales for the next 30 days:

Model 1: Basic Prophet Model

- Only used ds (date) and y (daily total sales).

- Provided a good general trend but had some deviations on high-spike days.

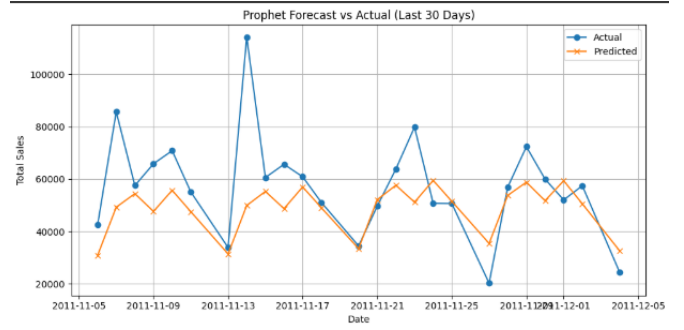


Fig. 6. Prophet forecast vs. actual sales for the last 30 days using the basic model with no regressors. While the trend is generally captured, the model struggles with sudden spikes.

Performance Metrics:

- MAE: 11,028.09
- RMSE: 14,278.88
- MAPE: 18.69%

These values indicate moderate forecast accuracy. MAE and RMSE give an average scale of error in currency units (Lee, 2021), while MAPE shows prediction deviation as a percentage (below 20% is acceptable).

Model 2: Improved Prophet with Regressors + Outlier Capping To improve prediction:

- Outliers in daily sales were capped using IQR.
- Added custom regressors like is_weekend, days_until_month_end, and known spike days.

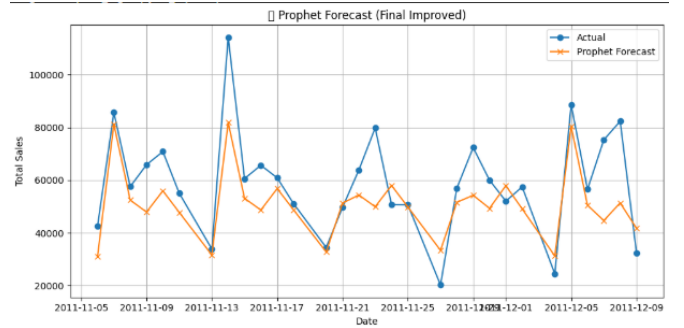


Fig. 7. Final improved Prophet forecast vs. actual sales. The inclusion of custom regressors and outlier capping significantly enhanced the model's ability to track peak and low periods more accurately.

Improved Metrics:

- MAE: 8,388.08
- RMSE: 11,610.79
- MAPE: 14.64%

The error rates improved significantly, with MAPE dropping by $\sim 4\%$, making this a more reliable model. The addition of regressors allowed the model to account for real-world periodic effects and anomalies.

TABLE I
MODEL EVALUATION SUMMARY

Model	MAE	RMSE	MAPE (%)	Notes
K-Means (Cluster Quality)	–	–	–	Silhouette: 0.4332, DB Index: 0.5340
Prophet (Basic)	11,028	14,278	18.69	Good baseline, misses on spike days
Prophet (Improved)	8,388	11,610	14.64	Best model with feature engineering

VI. DISCUSSION OF THE FINDINGS

The project managed to convert raw e-commerce transactional data into usable insights using systemized data cleaning, analysis, and modeling. Exploratory Data Analysis (EDA) established that there were a few popular products dominating the sales and customer's behavior assumed a long-tail behavior where most sales were in relatively low quantities and values. The clustering techniques recognized unique segments of customers through frequency and amount spent on orders; which determines targeted marketing. The forecasting models took into consideration overall sales trends and spikes in the seasons, and the improved performance was accomplished by the inclusion of the domain-specific features and the consideration of the outliers. The findings are in line that a thorough knowledge of business patterns is achieved through a combination of visual, statistical, and machine learning techniques.

VII. ANALYSIS OF THE FINDINGS

The findings demonstrate obvious structural trends to the data on online retail. Through data analysis, it was revealed that few products and few customers was responsible for most of the revenue and this was in accordance with the Pareto principle. Time-based analysis revealed predictable spikes during holidays with visualization tools making these patterns to be easily interpretable. The use of the clustering model provided viable customer profiling segmentation, justified by robust metrics of evaluation. At the same time, the forecasting models, and particularly enhanced Prophet version, yielded accurate sales prediction, greatly lowering error margins. Overall, the project shows how comprehensive approaches to data science can encourage smarter decisions in the e-commerce business.

VIII. CONCLUSION

This project was aimed at seeing how big data analytics could be used to reveal the customer purchasing behavior, and product trends in an e-commerce environment. The study used a systematic approach involving data cleaning, exploratory data analysis, customer segmentation based on K-means clustering, and sales forecasting as a result of the Prophet model from the rough transactional data. Based on key findings, a small portion of products as well as customers accounted for the majority of the revenue, customers' transactions were characterized by seasonal and behavioral patterns. The clustering approach effectively partitioned the customers using the purchasing behavior, while the improved forecasting model showed improved accuracy when domain-specific features and outlier treatment were applied. In general, the project proved

that the integration of statistical analysis, visualization, and machine learning models allow understanding the sales data within a detailed perspective and assist the business in making more informed decisions (Chen & Zhang, 2020; Taylor, 2021)

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